

Module 3: Risk and Risk Factors

100 Tutorial Topics: Algorithmic Bias, Fairness, Explainability, GenAI Risks

GARP RAI Exam Preparation

TOPIC: Why AI Risk Management is Challenging

KEY CHARACTERISTICS OF AI RISKS:

- **Pervasiveness:** AI is used across many domains - risk takes different forms depending on use case
- **High-Stakes Capabilities:** AI increasingly used in consequential settings - shortcomings have significant consequences
- **Rapid Development:** Capabilities advance faster than understanding of risks
- **Governance Challenges:** Regulatory responses are slow to develop and difficult to change

WHY IT MATTERS:

- Loan application algorithms profoundly affect applicants
- Medical diagnostic algorithms have long-lasting impact on patients
- Slow regulation allows harmful technologies to persist
- Normalization and entrenchment of harmful practices

INTERCONNECTED NATURE OF RISKS:

- Lack of transparency affects fairness and safety
- Algorithmic bias leads to reputational risks
- Risks reinforce each other - need holistic approach

EXAM TRAP: AI risks are **evolving, pervasive, and interconnected** - not static or isolated!

T2: Pervasiveness Challenge

TOPIC: Pervasiveness of AI and Risk Prediction Challenges

DEFINITION:

- AI is hugely pervasive in both range and variety of use
- Applications span many domains: finance, healthcare, justice, employment, etc.

WHY PERVASIVENESS POSES CHALLENGES:

- Risks can take on different forms depending on the use case
- A loan denial algorithm has different risks than a medical diagnostic algorithm
- Risk prediction must account for context
- No one-size-fits-all approach

EXAMPLES OF DIVERSE APPLICATIONS:

- **Credit Scoring:** Assessing loan applications
- **Medical Diagnostics:** Identifying diseases from images
- **Policing:** Predictive policing and bail decisions
- **Employment:** Resume screening and hiring
- **Education:** College admissions

EXAM TRAP: AI risks are **context-dependent** - must understand specific use case!

TOPIC: Consequences of AI in High-Stakes Settings

DEFINITION:

- AI technologies have increasingly large capabilities
- Used in increasingly high-stakes settings
- Technological shortcomings can have significant consequences for people

EXAMPLES OF HIGH-STAKES CONSEQUENCES:

- **Loan Applications:** Profound effect on applicant's financial well-being
- **Medical Diagnosis:** Long-lasting impact on patient's life
- **Bail Decisions:** Affect personal freedom
- **Hiring Decisions:** Impact career trajectories

IMPLICATIONS:

- Algorithmic errors can have profound impacts on people's lives
- Need for robust testing and validation
- Human oversight essential

EXAM TRAP: High-stakes AI = significant consequences for individuals!

TOPIC: Challenges from Rapid AI Development

DEFINITION:

- AI development is rapid
- Capabilities advance faster than our understanding of risks

CHALLENGES CREATED:

- **Impact Prediction:** Hard to predict future capabilities and risks
- **Risk Measurement:** Metrics may not capture new risks
- **Risk Mitigation:** Solutions may lag behind new developments
- **Governance:** Difficult to establish robust mechanisms

REGULATORY CHALLENGES:

- Regulatory responses are slow to develop
- Difficult to change once established
- Slow regulation allows harmful technologies to persist
- Normalization and entrenchment of harmful practices

EXAM TRAP: Rapid development = governance and regulation challenges!

TOPIC: How AI Risks are Interconnected

DEFINITION:

- Risks from AI are often interconnected
- Addressing one risk may affect others

EXAMPLES OF INTERCONNECTIONS:

- **Transparency → Fairness:** Lack of transparency affects fairness assessment
- **Bias → Reputation:** Algorithmic bias leads to reputational risks
- **Safety → Fairness:** Safety failures can reveal fairness issues
- **Explainability → Accountability:** Lack of explainability creates accountability gaps

IMPLICATIONS:

- Need holistic approach to risk management
- Cannot address risks in isolation
- Must consider entire AI lifecycle

EXAM TRAP: AI risks are **interconnected and mutually reinforcing!**

T6: Algorithmic Bias Definition

TOPIC: Algorithmic Bias - Definition and Types

DEFINITION:

- Algorithmic bias is a systematic deviation of an algorithm's output, performance, or impact relative to some norm, aim, standard, or baseline
- Systematic (not random)
- Can affect output, performance, or impact
- Relative to some norm, aim, standard, or baseline

TWO TYPES:

- **Explicit Bias:** Developer intentionally projects own biases into algorithm
- **Implicit Bias:** Bias creeps in undetected through design or data

WHY IT MATTERS:

- Algorithms can exhibit bias and create unfair results
- Systematic nature means bias affects many people consistently
- Regulators increasingly demand fair and non-discriminatory AI

EXAM TRAP: Algorithmic bias = **systematic deviation from a norm!**

T7: Explicit vs Implicit Bias

TOPIC: Distinguishing Between Explicit and Implicit Bias

EXPLICIT BIAS:

- Developer intentionally projects own biases into algorithm
- Leads to biased outputs
- Intentional discrimination

IMPLICIT BIAS:

- Bias creeps in undetected
- Through algorithm's design or training data
- Unintended
- Often harder to detect and correct

EXAMPLES:

- **Explicit:** Deliberately excluding certain groups
- **Implicit:** Historical data reflecting past discrimination

KEY INSIGHT:

- Both types can lead to unfair outcomes
- Implicit bias is more common in practice
- Need to check for both types

EXAM TRAP: Explicit = intentional; Implicit = unintended!

T8: Individual Fairness

TOPIC: Individual Fairness - Treating Like Cases Alike

DEFINITION:

- Based on Aristotelian doctrine of "treating like cases alike"
- Similarly situated individuals should be treated equally
- Focuses on individuals, not groups

KEY CHARACTERISTICS:

- Treating applicants alike in all relevant aspects
- Avoids direct discrimination on protected characteristics
- Requires agreement on what counts as "alike"

LIMITATIONS:

- What counts as "alike" is often contentious
- Should we consider hardships overcome?
- How to translate resilience into numbers?
- Different people have different views

EXAMPLE (College Admission):

- If grades are the only relevant metric, select highest grades
- But students start from different positions
- Should socioeconomic factors matter?

EXAM TRAP: Individual fairness = treating like cases alike!

TOPIC: Group Fairness - Statistical Differences Across Groups

DEFINITION:

- Does not consider individuals
- Looks at statistical differences across groups
- Groups share protected characteristics

KEY CHARACTERISTICS:

- Examines whether groups are disadvantaged
- Examples: admission rates by gender, loan approvals by race
- Two categories:
 - Performance equality
 - Output distribution

MAIN MEASURES:

- **Demographic Parity:** Equal distribution of predictions
- **Predictive Rate Parity:** Equal precision
- **Equal Opportunity:** Equal sensitivity

EXAM TRAP: Group fairness = statistical differences across groups!

TOPIC: Fairness Challenges in College Admission

SCENARIO:

- College uses algorithm CollegeAd
- Analyzes test scores, transcripts, writing samples
- Outputs hierarchy of most promising students

FAIRNESS QUESTIONS:

- What counts as "alike"?
- Should we consider hardships overcome?
- How to translate resilience into numbers?
- Should socioeconomic factors matter?

LIMITATIONS OF INDIVIDUAL FAIRNESS:

- Intuitive but requires agreement on "alike"
- Often contentious
- Designers must be explicit about relevant factors
- Factors may vary with context

EXAM TRAP: Individual fairness requires **agreement on what is "alike"**!

TOPIC: Demographic Parity - Definition and Formula

DEFINITION:

- Distribution of predictions is identical across subpopulations
- Admission rate same across groups
- Loan approval rate same across groups

FORMULA:

$$P(\hat{Y} = 1|A = 0) = P(\hat{Y} = 1|A = 1)$$

where:

- $\hat{Y} \in \{0, 1\}$: Predicted outcome
- $A \in \{0, 1\}$: Protected attribute/group membership

KEY CHARACTERISTIC:

- Entirely independent of true value
- Does not consider if predictions are correct
- Only if predictions are equally distributed

EXAM TRAP: Demographic parity = equal distribution of predictions!

T12: Demographic Parity - Drawbacks

TOPIC: Drawbacks of Demographic Parity

MAJOR DRAWBACK:

- Does not consider quality/accuracy of predictions
- Can be satisfied even if all predictions are wrong
- Example: Algorithm performs worse on subpopulation, labels many unqualified as qualified
- Still satisfies demographic parity if proportion is same

WHEN INAPPROPRIATE:

- Base rates significantly differ between groups
- Differences are relevant to decision
- Medical context: quality of prediction is primary concern
- Credit risk: relevant factors correlated with protected attributes

WHEN USEFUL:

- When true value not directly accessible
- Credit scoring: time lapse before outcome observed
- Fraud detection: true fraud cases limited

EXAM TRAP: Demographic parity ignores **prediction accuracy!**

TOPIC: Predictive Rate Parity - Definition and Formula

DEFINITION:

- Precision is equal across subpopulations
- Among those predicted positive, proportion correctly identified is same
- Also called Positive Predictive Value (PPV) parity

FORMULA:

$$P(Y = 1 | \hat{Y} = 1, A = 0) = P(Y = 1 | \hat{Y} = 1, A = 1)$$

IN TERMS OF PRECISION:

$$\text{Precision}_0 = \text{Precision}_1$$

where $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$

INTUITIVE APPEAL:

- Avoid stark differences in false positives across groups
- Ensure proportion of accepted applicants who succeed is same
- Prevent academic potential being overestimated in one group

EXAM TRAP: Predictive rate parity = equal precision!

T14: Predictive Rate Parity - Drawbacks

TOPIC: Drawbacks of Predictive Rate Parity

MAJOR DRAWBACK:

- Does not account for differences in base rates across groups
- If base rates differ significantly, achieving parity is challenging
- Can have adverse effects on algorithmic performance

WHY CHALLENGING:

- Base rates = general prevalence within population
- Disease prevalence may differ across groups
- Default rates may differ across demographics
- Achieving parity may require sacrificing accuracy

EXAMPLES:

- Medical diagnostics: disease rates differ
- Credit scoring: default rates differ
- College admissions: qualification rates differ

EXAM TRAP: Predictive rate parity struggles with **different base rates!**

TOPIC: Equal Opportunity - Definition and Formula

DEFINITION:

- Sensitivity (True Positive Rate) is equal across groups
- Individuals of each group have equal chance of being correctly identified as positive
- Important when positive cases must be recognized

FORMULA:

$$P(\hat{Y} = 1|Y = 1, A = 0) = P(\hat{Y} = 1|Y = 1, A = 1)$$

IN TERMS OF SENSITIVITY:

$$\text{Sensitivity}_0 = \text{Sensitivity}_1$$

where $\text{Sensitivity} = \text{TP}/(\text{TP} + \text{FN})$

EXAMPLE (Medical Diagnostics):

- Patients with disease equally likely to be diagnosed correctly
- Regardless of demographic group
- Don't miss disease in any group

EXAM TRAP: Equal opportunity = equal true positive rate (sensitivity)!

TOPIC: Drawbacks of Equal Opportunity

MAJOR DRAWBACK:

- Requires access to true value of target
- Can be difficult to achieve with stark base rate differences
- Difficult to achieve fairness without compromising other performance aspects

RELATED CONCEPT - EQUALIZED ODDS:

- More restrictive version of equal opportunity
- Requires equality of sensitivity AND specificity across groups
- $P(\hat{Y} = 1 | Y = 1, A = 0) = P(\hat{Y} = 1 | Y = 1, A = 1)$
- $P(\hat{Y} = 1 | Y = 0, A = 0) = P(\hat{Y} = 1 | Y = 0, A = 1)$

WHEN USEFUL:

- Ensure same proportion of qualified candidates admitted
- Ensure don't miss disease in any group

EXAM TRAP: Equal opportunity requires **access to true values!**

T17: Fairness Impossibility Theorem

TOPIC: Fairness Impossibility Theorem and Trade-offs

DEFINITION:

- Mathematical theorem demonstrates impossibility
- When base rates differ between populations (almost always)
- Cannot satisfy demographic parity, predictive rate parity, and equal opportunity simultaneously

IMPLICATIONS:

- Cannot achieve fairness on all dimensions
- Must deliberate on kind of group fairness to achieve
- Trade-offs are unavoidable
- Must choose based on context

PERFORMANCE TRADE-OFFS:

- Balancing different performance measures is non-trivial
- Example: Fraud detection - sensitivity vs. false-positive rate
- Block all transactions: maximum sensitivity, high false-positive rate
- Label all as non-fraudulent: minimum false-positive rate, low sensitivity

EXAM TRAP: Cannot satisfy **all fairness measures** simultaneously!

TOPIC: Summary of Group Fairness Measures

THREE KEY MEASURES:

Measure	Question Answered
Demographic Parity	"Is the likelihood of a positive prediction the same across groups?"
Predictive Rate Parity	"Is the likelihood of a true positive prediction the same across groups?"
Equal Opportunity	"Is the likelihood of identifying actual positives as positive the same across groups?"

KEY CHARACTERISTICS:

- **Demographic Parity:** Independent of correctness
- **Predictive Rate Parity:** Focuses on precision
- **Equal Opportunity:** Focuses on sensitivity

EXAM TRAP: Know the **question each measure answers!**

TOPIC: Precision in Classification

DEFINITION:

- Proportion of positive predictions that are correct
- Also known as Positive Predictive Value (PPV)

FORMULA:

$$\text{Precision} = \frac{TP}{TP + FP}$$

INTERPRETATION:

- "Of those predicted positive, how many were actually positive?"
- High precision: Few false positives
- Low precision: Many false positives

EXAMPLE (Credit Scoring):

- Proportion of applicants predicted to default who actually default
- High precision: Algorithm effectively identifies likely defaulters
- Low precision: Many who would repay erroneously labeled high-risk

EXAM TRAP: Precision = $TP/(TP+FP)$!

TOPIC: Sensitivity (Recall) in Classification

DEFINITION:

- Proportion of actual positive instances correctly predicted
- Also known as True Positive Rate (TPR) or Recall

FORMULA:

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

INTERPRETATION:

- "Of those actually positive, how many did we catch?"
- Useful when important not to miss a positive
- Maximizing sensitivity requires minimizing false negatives

RELATED CONCEPT:

- **Specificity**: Proportion of actual negatives identified
- $\text{Specificity} = \frac{TN}{TN + FP}$

EXAM TRAP: Sensitivity = $TP / (TP + FN)$!

TOPIC: Accuracy in Classification

DEFINITION:

- Fraction of predictions that are correct
- Intuitive but often misleading

FORMULA:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

PROBLEM WITH ACCURACY:

- Misleading with unbalanced data
- Example: 100 applicants, 10 suitable, 90 unsuitable
- Algorithm classifies all as unsuitable: Accuracy = 90%
- But provides no useful information

WHEN TO USE:

- Only when classes are balanced
- Or when costs of errors are equal

EXAM TRAP: Accuracy is misleading with unbalanced data!

TOPIC: False Positive Rate (FPR)

DEFINITION:

- Proportion of false positives out of all actual negative cases

FORMULA:

$$FPR = \frac{FP}{FP + TN}$$

RELATIONSHIP TO SPECIFICITY:

$$FPR = 1 - \text{Specificity}$$

INTERPRETATION:

- "Of all actual negatives, how many were incorrectly flagged positive?"
- Important for fairness analysis
- Used in ROC curves

EXAM TRAP: $FPR = FP/(FP+TN)$!

TOPIC: Confusion Matrix

DEFINITION:

- Cross-tabulation of predicted vs. actual outcomes
- Foundation for all classification metrics

STRUCTURE:

	Predicted Neg	Predicted Pos
Actual Neg	TN	FP
Actual Pos	FN	TP

COMPONENTS:

- **True Negative (TN)**: Correctly predicted negative
- **False Positive (FP)**: Incorrectly predicted positive
- **False Negative (FN)**: Incorrectly predicted negative
- **True Positive (TP)**: Correctly predicted positive

EXAM TRAP: Confusion matrix = TP, TN, FP, FN!

TOPIC: Problem Specification and Feature Selection

DEFINITION:

- During problem specification, purpose and design of algorithm are explicitly stated
- How a problem is specified affects algorithmic performance and fairness

KEY ISSUES:

- Choice of criteria affects who applies and who gets hired
- Domain expertise is necessary for good problem specification
- Translation to measurable features requires proxies
- Some things harder to quantify than others

EXAMPLE (Hiring Algorithm):

- Company wants to improve recruitment using HireMe
- Scans CVs, outputs shortlist of best applicants
- Need to specify what is a "good applicant"
- Productivity, work ethic, team player, retention

EXAM TRAP: Problem specification = **critical for fairness!**

T25: Proxy Variables

TOPIC: Proxy Variables in Algorithmic Design

DEFINITION:

- Variables used as stand-ins for concepts hard to measure directly
- Judged to be strongly correlated with the criterion

WHY PROXIES ARE USED:

- Cannot easily measure productivity directly
- Can measure sales made within one year
- Simpler measurable construct

RISKS:

- Reduces complex concepts
- May introduce bias
- Example: Total sales misses that women more likely to work part-time
- Lower total sales rate for women

EXAMPLE:

- Productivity measured by sales
- Teamwork harder to quantify
- Skipping features may hurt candidates
- If teamwork area where women outperform men

EXAM TRAP: Proxies are **imperfect stand-ins!**

T26: Amazon Recruiting Tool

TOPIC: Amazon Recruiting Tool Example

SCENARIO:

- Amazon attempted to create unbiased recruiting tool
- Provided algorithm with previous applicants' CVs
- Target variable: whether hired
- "Good" candidate = resembles previous hires

OUTCOME:

- Algorithm systematically downgraded women's CVs
- Even identical CVs rated lower for women
- Amazon abandoned the project

LESSON:

- Learning from historical data perpetuates biases
- Past discrimination reflected in training data
- Cannot simply rely on past patterns

EXAM TRAP: Amazon's tool **systematically discriminated against women!**

TOPIC: Fairness Through Unawareness - Does It Work?

DEFINITION:

- Intuitive approach: Remove protected characteristics from inputs
- Assumes this prevents discrimination
- Embedded in most legal doctrines

PROBLEM:

- Correlated variables carry protected information
- Removing protected variable doesn't eliminate its influence
- Correlated variables weighted more strongly

EXAMPLE (FTA Prediction):

- JusticeGuard predicts failure to appear in court
- Protected group and prior convictions correlated
- Remove protected variable
- Prior conviction coefficient increases from 54% to 72%

EXAM TRAP: Fairness through unawareness **does NOT** eliminate bias!

TOPIC: FTA Prediction - Mathematical Illustration

INITIAL MODEL (with protected characteristic):

$$\hat{Y} = 0.034 + 0.543X_{prior} + 0.332X_{prot}$$

- Prior offense: 54% increase in risk
- Protected group: 33% increase in risk

REMOVE PROTECTED CHARACTERISTIC:

$$\hat{Y} = \gamma_0 + \gamma_1 X_{prior}$$

RESULT:

$$\hat{Y} = 0.10 + 0.72X_{prior}$$

- Prior now increases risk by 72% (not 54%)
- Coefficient contaminated by correlation
- Algorithm still indirectly discriminates

EXAM TRAP: Removing protected variable **can make bias worse!**

TOPIC: Historical Bias in Training Data

DEFINITION:

- Supervised learning relies on historical data
- Historical data reflects past biases and discrimination
- Algorithm learns and may exacerbate these patterns

EXAMPLE:

- Hiring algorithm trained on past hiring data
- Women historically less likely selected for certain roles
- Algorithm repeats same patterns
- May exacerbate them

CONSEQUENCE:

- Systematic bias perpetuated
- Discrimination becomes automated
- Hard to detect and correct

EXAM TRAP: Historical bias = past discrimination reflected in data!

TOPIC: Feedback Loops in AI Systems

DEFINITION:

- Algorithmic predictions influence future data
- Biased predictions create more biased data
- Amplifies existing discrimination

HOW IT WORKS:

- 1 Algorithm makes biased predictions
- 2 Decisions based on predictions create new data
- 3 New data reflects biased decisions
- 4 Algorithm learns from biased data
- 5 Bias amplified

EXAMPLE:

- Hiring algorithm learns from biased outcomes
- Fewer women hired
- Algorithm sees fewer women in successful hires
- Future predictions even more biased

EXAM TRAP: Feedback loops **amplify** and **perpetuate** bias!

TOPIC: Representation Bias in Data Collection

DEFINITION:

- Collected sample not representative of target population
- Results in problematic model specifications

EXAMPLES:

- Facial recognition trained mostly on white faces
- Medical testing with predominantly white/European samples
- Algorithms perform better on dominant groups

INTERSECTIONAL BIAS:

- Performance particularly low for intersectional groups
- Example: Women of color in facial recognition
- Multiple dimensions of underrepresentation

EXAM TRAP: Representation bias = **non-representative sample!**

TOPIC: Measurement Bias in Data Collection

DEFINITION:

- Measurement methods not consistent across sample
- Can favor certain groups

EXAMPLE:

- Doctor spends more time with wealthy patients
- More thorough evaluations
- Data on early-stage disease discovery biased
- Favor of wealthy group

CONSEQUENCE:

- Biased training data
- Algorithm learns incorrect patterns
- Systematic errors for certain groups

EXAM TRAP: Measurement bias = inconsistent measurement across groups!

TOPIC: Data Composition Bias

DEFINITION:

- Minority subgroups underrepresented in data
- Algorithms underperform for these groups
- Statistical nature of learning delivers statistically ideal output

EXAMPLE:

- Medical training set: 3% pregnant women
- Pregnant women require different treatment
- Algorithm significantly underperforms for this group
- Fewer data points to learn from

SOLUTION:

- Balance datasets
- Relevant minority subgroups occupy similar proportion
- Ensures algorithm recognizes minority groups as statistically relevant

EXAM TRAP: Data composition bias = **underrepresented minority subgroups!**

TOPIC: Balancing Datasets

DEFINITION:

- Ensuring relevant minority subgroups occupy similar proportion as majority groups
- Different from proportional representation

EXAMPLE:

- **Proportional:** 10% triangles, 90% circles → dataset same
- **Balanced:** Increase proportion of triangles in dataset
- Male/female 50% even if historically 20% female

METHODS:

- **Over-sampling:** Add more data from minority group
- **Under-sampling:** Remove data from majority group
- **Combination:** Both methods
- **Synthetic data:** Generate new data points

EXAM TRAP: Balancing = **over-representing minorities** for algorithm performance!

TOPIC: Bootstrapping for Data Balancing

DEFINITION:

- Resampling datapoints from existing dataset
- Adding back into dataset

THREE METHODS:

- 1 **Over-sampling:** Datapoints from minority group
- 2 **Under-sampling:** Remove datapoints from majority group
- 3 **Combination:** Both methods

CHALLENGES:

- Creating synthetic data without compromising generalization
- Taking out datapoints works only if datasets are large enough
- Rare cases especially challenging (fraud $\leq 1\%$)

ADDITIONAL METHODS:

- Ensemble methods: Use multiple models
- Cost function: Higher penalty for minority class misclassification

EXAM TRAP: Bootstrapping includes **over-sampling AND under-sampling!**

TOPIC: Intersectional Bias

DEFINITION:

- Bias affecting groups at intersection of multiple underrepresented subgroups
- Performance particularly low for intersectional groups

EXAMPLE:

- Women of color in facial recognition
- Multiple dimensions of underrepresentation
- Harder to detect and correct

WHY IT MATTERS:

- Standard fairness metrics may miss intersectional bias
- Need to analyze multiple dimensions
- Particularly harmful for affected individuals

EXAM TRAP: Intersectional bias = **multiple underrepresented dimensions combined!**

TOPIC: Word Embeddings Bias

DEFINITION:

- Preprocessing technique for NLP
- Transforms words into high-dimensional vectors
- Captures meaning through relationships to other words

EXAMPLES:

- man is to woman as king is to queen
- man is to computer programmer as woman is to homemaker
- Contains stereotypes

CONSEQUENCE:

- NLP algorithms incorporate stereotypes
- May perpetuate harmful biases
- Difficult to detect and correct

EXAM TRAP: Word embeddings can contain **harmful stereotypes!**

TOPIC: Objective Function Bias

DEFINITION:

- Choice of objective function introduces value judgments
- What algorithm optimizes for makes ethically significant difference

KEY QUESTIONS:

- Minimize prediction errors (focus on performance)?
- Or focus on how errors are distributed (focus on error distribution)?
- Choice depends on context

EXAMPLE (Cancer Detection):

- **Skin cancer:** Err on side of caution
- **Brain cancer:** Treatment highly invasive
- Different trade-offs for false positives/negatives
- Context matters for objective function

EXAM TRAP: Objective function choice = ethical significance!

TOPIC: Benchmark Dataset Bias

DEFINITION:

- Algorithms tested against publicly available benchmark datasets
- May not be representative of intended use population
- Not necessarily good indicator of model quality

PROBLEM:

- Good performance on benchmark doesn't guarantee real-world performance
- Can mask biases that appear in deployment
- Need domain-specific testing

EXAM TRAP: Benchmark datasets may **not reflect real-world populations!**

TOPIC: Deployment Context Bias

DEFINITION:

- Algorithms used in different contexts than designed for
- Loss of performance or other challenges
- Applies to different contexts and outdated data

EXAMPLES:

- Recidivism prediction for sentence length
- Values of users differ from developers
- Third-party algorithms may not align with company values

SOLUTIONS:

- Human oversight of data collection
- Oversight of development
- Oversight of deployment
- Regular auditing

EXAM TRAP: Deployment context = critical for performance and fairness!

TOPIC: Importance of Human Oversight

DEFINITION:

- Ensuring safe and fair algorithms requires human oversight
- Throughout AI lifecycle

AREAS REQUIRING OVERSIGHT:

- Data collection process
- Development of algorithm
- Deployment
- Regular auditing

WHY IT MATTERS:

- Algorithms deployed in consequential domains
- Need accountability
- Regular auditing becomes important
- Failures can have severe consequences

EXAM TRAP: Human oversight is **essential throughout AI lifecycle!**

T42: Sources of Unfairness Summary

TOPIC: Summary of Sources of Unfairness

MAJOR SOURCES:

- ➊ **Problem Specification:**
 - Choice of criteria
 - Translation to measurable features
 - Proxies
- ➋ **Data Collection:**
 - Historical bias
 - Representation bias
 - Measurement bias
- ➌ **Data Composition:**
 - Imbalanced datasets
 - Statistical minorities
- ➍ **Model Development:**
 - Preprocessing techniques
 - Choice of objective function
 - Benchmark testing
- ➎ **Deployment:**
 - Context mismatch
 - Value misalignment

EXAM TRAP: Multiple sources - need **holistic approach!**

TOPIC: Explainability in AI

DEFINITION:

- Capacity to explain in understandable terms how AI system makes decisions
- Often after the fact (ex-post explainability)

WHY IT MATTERS:

- Inability to explain leads to accountability gaps
- Erosion of trust among users and stakeholders
- Essential in consequential domains (healthcare, finance)

TWO ASPECTS:

- **Philosophical:** What counts as a good explanation?
- **Technical:** How to obtain good explanations?

EXAM TRAP: Explainability = explaining decisions after the fact!

TOPIC: Interpretability in AI

DEFINITION:

- Degree to which human can comprehend and predict model's behavior
- Built-in mechanisms to understand inherently
- How inputs affect outputs

INHERENTLY INTERPRETABLE MODELS:

- Decision trees: Can follow logic
- Linear regression: Coefficients explain impact
- Rule-based systems: Explicit rules

BLACK-BOX MODELS:

- Neural networks: Complex architecture
- Ensemble methods: Hard to trace logic
- Less interpretable

EXAM TRAP: Interpretability = **inherently understandable models!**

TOPIC: Transparency in AI

DEFINITION:

- Openness and accessibility of AI system's design, algorithms, data, and decision-making processes
- Builds trust, facilitates collaboration
- Enables oversight and governance

WHY IT MATTERS:

- Essential for accountability
- Enables alignment with ethical standards
- Aligns with organizational values
- Meets regulatory requirements

CHALLENGES:

- Security concerns (spam filters)
- Proprietary interests
- Specialized knowledge required

EXAM TRAP: Transparency = openness and accessibility!

TOPIC: The Black-Box Problem

DEFINITION:

- Model too complex to be interpreted
- Humans cannot understand intrinsic architecture
- Leads to explainability issues

EXAMPLES:

- Neural networks: Millions of parameters
- Deep learning models: Complex interactions
- Ensemble methods: Hard to trace

CONSEQUENCES:

- Understanding decision-making process difficult
- If not impossible
- Accountability gaps
- Trust issues

EXAM TRAP: Black-box problem = models too complex to interpret!

TOPIC: Feature Importance Scores

DEFINITION:

- Prominent technique in Explainable AI (XAI)
- Ranks importance of input features for outcomes

EXAMPLE:

- Credit-scoring AI system
- Reveals which factors influence credit score
- Income, credit history, etc.

BENEFITS:

- Scrutinize how model makes decisions
- Identify potential biases or errors
- Analyze impact of input features

EXAM TRAP: Feature importance = ranks feature influence!

TOPIC: Shapley Values in XAI

DEFINITION:

- Calculate contribution of each input feature to model output
- Measure marginal effect of including each feature
- Across all possible feature combinations

BASED ON GAME THEORY:

- Fair attribution of contributions
- Considers all possible coalitions
- Provides consistent explanations

APPLICATIONS:

- Understanding model decisions
- Identifying important features
- Regulatory explanations

EXAM TRAP: Shapley values = feature contribution calculation!

TOPIC: LUCID in XAI

DEFINITION:

- LUCID = Locating Unfairness through Canonical Inverse Design
- Works backward from algorithmic output
- Creates distribution over input space

PROCESS:

- Start from particular output (e.g., "loan granted")
- Create distribution over inputs
- Reveals model's internal logic
- Helps locate unfairness

BENEFITS:

- Understanding internal logic
- Identifying sources of unfairness
- Informing mitigation strategies

EXAM TRAP: LUCID works **backward from output to reveal logic!**

TOPIC: Surrogate Models in XAI

DEFINITION:

- Simpler, more interpretable models
- Approximate behavior of complex AI system

EXAMPLE:

- LIME (Local Interpretable Model-agnostic Explanations)
- Approximates complex model's prediction locally
- Uses interpretable model (linear regression, decision tree)
- Model-agnostic

BENEFITS:

- Enhance interpretability
- Approximate complex model locally
- Can be quite accurate locally

EXAM TRAP: Surrogate models are **simpler approximations!**

TOPIC: Challenges with Explainable AI

KEY CHALLENGES:

1 Oversimplification:

- Complex decisions oversimplified
- Can distort understanding
- Lead to misinterpretation

2 Computational Cost:

- Several XAI techniques computationally intensive
- More costly to develop and deploy

3 Ex-Post Nature:

- Used to understand models after training
- Not built into model design

TRADE-OFF:

- Complex models often achieve higher accuracy
- But are less interpretable
- Must weigh both factors in context

EXAM TRAP: XAI can oversimplify and distort understanding!

TOPIC: Secrecy as a Form of Opaqueness

DEFINITION:

- Individual not aware of being subject to algorithmic decision making
- Data extracted without knowledge

EXAMPLE:

- Algorithm extracts personal data from social media
- Clicks, buys, website visits
- Extrapolates profile
- Makes recommendations for advertisements
- Without user's knowledge

VIOLATION:

- Transparency and consent
- User autonomy
- Privacy

EXAM TRAP: Secrecy = unawareness of algorithmic decision making!

TOPIC: Confidentiality as a Form of Opaqueness

DEFINITION:

- Affected parties aware of being subject to algorithmic decision making
- But don't have knowledge of how decisions are made
- Don't understand reasons for outcomes

MAIN CONCERN:

- Recent discourse on algorithmic decision making
- Not unique to algorithms
- Secret decision-making in general

EXAMPLE:

- Automatically denied loan
- No way to find out why
- Frustration and distrust

EXAM TRAP: Confidentiality = **opaque** decision-making process!

TOPIC: Specialized Knowledge as a Form of Opaqueness

DEFINITION:

- Understanding algorithm requires specialized training
- Even with access to source code and data
- May not be able to make sense of it

CHARACTERISTICS:

- Not unique to AI systems
- Results from lack of preexisting knowledge
- Creates unequal access to understanding

IMPLICATIONS:

- Need for expert intermediaries
- Challenges for meaningful consent
- Accountability gaps

EXAM TRAP: Specialized knowledge = requires training to understand!

TOPIC: Valid Reasons for Algorithmic Opacity

SECURITY:

- Making code public enables circumvention
- Example: Spam filters would be easier to bypass
- Risk of fraud, scams, cyber-attacks

PROPRIETARY INTERESTS:

- Trade secrets
- Maintaining competitive advantage
- Algorithm as intellectual property

BALANCING ACT:

- Companies should obtain access for risk assessments
- Even with third-party algorithms
- Provide accountability

EXAM TRAP: Legitimate reasons include [security](#) and [proprietary interests](#)!

TOPIC: Human Autonomy in the Age of AI

DEFINITION:

- Ability to hold beliefs, make, and execute decisions of our own
- Two dimensions

TWO DIMENSIONS:

- **Internal:**
 - Values, beliefs, decisions that are our own
 - Not product of distorting external factors
 - Not manipulation
- **External:**
 - Ability to execute decisions
 - Freedom and opportunity to do so

MISCONCEPTION:

- Delegating tasks to AI doesn't automatically lose autonomy
- Routine outsourcing doesn't reduce autonomy
- As long as adequate control and human oversight

EXAM TRAP: Autonomy = own beliefs and decisions + ability to execute!

TOPIC: Cambridge Analytica Scandal

SCENARIO:

- Collected Facebook user data
- Users completed personality tests
- Inferred personality profiles from online behavior
- Sent psychologically tailored political advertising

DEMONSTRATED:

- Big data can be used to manipulate voter behavior
- Potential for political manipulation
- Need for robust data privacy and security

REPUTATIONAL IMPACT:

- Facebook faced reputational damage
- Brand value loss
- Public trust erosion

EXAM TRAP: Cambridge Analytica = **political manipulation through data!**

TOPIC: Facebook Emotional Contagion Experiment

SCENARIO:

- Researchers changed users' timelines
- Displayed predominantly positive or negative posts
- Users' own posts became more positive/negative
- No consent from users

DEMONSTRATED:

- Power of recommendation algorithms
- Ability to shape experience
- Potential to manipulate

PUBLIC OUTCRY:

- Users did not consent
- Violation of trust
- Ethical concerns

EXAM TRAP: Experiment showed [AI can manipulate emotional state!](#)

TOPIC: Preventing AI Manipulation

APPROACHES:

- **Human Oversight:**
 - Throughout development process
 - Meaningful human control
- **Regulatory Oversight:**
 - EU AI Act
 - Transparency obligations
 - Prohibiting subliminal techniques

CHALLENGES:

- Ambiguous notion of manipulation
- Difficult to operationalize
- Hard to make measurable

BALANCE:

- Benefits of personalization and targeting
- vs. risks of manipulation
- Need adequate oversight

EXAM TRAP: Prevention requires **human and regulatory oversight!**

T60: Uber Crash Example

TOPIC: Uber Driverless Vehicle Crash

SCENARIO:

- Uber driverless vehicle in Tempe, Arizona
- Fatally injured a pedestrian
- Multiple failures

FAILURES:

- **System Failure:**
 - Vehicle's systems failed to recognize pedestrian in time
- **Deactivated Safety:**
 - Some safety mechanisms were deactivated
- **Human Factor:**
 - Safety driver distracted
 - Did not react quickly enough
 - Result of automation bias

LESSONS:

- Risk of over-dependence on AI
- Need for robust testing and validation
- Comprehensive testing protocols

EXAM TRAP: Uber crash = risks of over-reliance on AI!

TOPIC: Automation Bias

DEFINITION:

- Tendency of humans to over-rely on automated systems
- Leading to complacency and reduced vigilance

CONSEQUENCES:

- Errors of oversight
- Especially problematic in critical situations
- Skill atrophy

SKILL ATROPHY:

- Prolonged reliance on automation
- Decay in essential skills
- Operators become passive observers
- From active decision makers to passive

EXAM TRAP: Automation bias = **over-reliance on automated systems!**

TOPIC: Mitigating Automation Bias

MITIGATION STRATEGIES:

1 Inform Users:

- Risks of automation bias
- Limitations of AI

2 Design Mechanisms:

- Require periodic human input
- Require verification
- Keep operators engaged and alert

EXAMPLE:

- Eye tracking technology
- Combined with alerts
- Ensure meaningful human control
- Uber could have used this

EXAM TRAP: Mitigation = user education + periodic human verification!

TOPIC: AI in Mental Health Support

APPLICATIONS:

- Chatbots
- Virtual therapists
- Offer accessibility and anonymity

CONCERNS:

- Adequacy in handling complex emotional issues
- AI has no empathy
- Empathy crucial for mental health support
- Positive effects dwindle when patients know AI-generated

GOLDEN RULE:

- Just because we can does not mean we should
- Reflect on which issues lend themselves to automation
- Some should stay in human hands

EXAM TRAP: AI lacks **genuine empathy** - critical for mental health!

TOPIC: Golden Rules for AI Development

RULE 1:

- "Just because we can does not mean we should"
- Reflect carefully on which issues lend themselves to automation
- Some issues should stay in human hands
- Until we have found ways to overcome inadequacies

RULE 2:

- Involve diverse stakeholders
- Domain experts, wider public, ethicists, policy makers
- For AI applications in critical situations
- Ensures technology is ethical, safe, appropriate

RULE 3:

- Developers and users must be transparent and accountable
- Clear lines of responsibility
- In case of failures or adverse outcomes

EXAM TRAP: Golden rules = reflection, inclusion, transparency!

TOPIC: Reputational Risks from AI

DEFINITION:

- Companies expose themselves to novel risks
- When use or output of algorithms goes against prevalent norms and values
- Reputational damage

THREE MAIN CAUSES:

1 Privacy Breaches:

- Cambridge Analytica example
- Mishandling user data

2 Algorithmic Bias:

- Dutch child benefits scandal
- Unfair algorithms

3 Lack of Explainability:

- Customers denied without reasons
- No way to find out why

EXAM TRAP: Three main causes = **privacy**, **bias**, **explainability**!

TOPIC: Dutch Child Benefits Scandal

SCENARIO:

- Fraud-detection algorithm deployed by Dutch government
- Falsely denied and reclaimed child benefits
- Thousands of Dutch citizens affected
- Dire consequences for families

OUTCOME:

- Dutch prime minister resigned
- Major political scandal
- Reputational damage to government

LESSONS:

- Algorithmic bias can have severe consequences
- Need for oversight and accountability
- Importance of fairness in government algorithms

EXAM TRAP: Dutch scandal = [algorithmic bias with severe consequences!](#)

TOPIC: Types of AI-Related Criticism

TWO CATEGORIES:

1 Competence Criticism:

- Perception that company lacks expertise
- Insufficient capacities for safe/fair algorithms
- Technical failings, unfairness, lack of explainability
- Data not stored safely

2 Value Alignment Criticism:

- Company fully aware of risks but proceeded
- Facebook emotional contagion experiment
- Violation of user trust and expectations
- Lack of due diligence

OVERLAP:

- Distinction not always clear-cut
- Data breach can be incompetence or misaligned values
- Perception may be both

EXAM TRAP: Competence = lack of expertise; Value alignment = misaligned values!

TOPIC: Mitigating Reputational Risks

STRATEGIES:

- ① **Continuous Monitoring and Evaluation:**
 - Identify risks proactively
 - Organizational governance mechanisms
 - Regular monitoring and evaluation
- ② **Ensuring Privacy and Security:**
 - Comply with GDPR
 - Secure data storage, handling, processing
 - Transparent data usage policies
- ③ **Ensuring Algorithmic Fairness:**
 - Train on diverse datasets
 - Regular audits for discriminatory patterns
- ④ **Promoting Transparency and Explainability:**
 - XAI techniques
 - Customers aware of algorithm use
 - Builds trust and accountability

EXAM TRAP: Mitigation = monitoring, privacy, fairness, transparency!

TOPIC: Demonstrating Will to Change After Incidents

DEFINITION:

- Once incident has happened, limit reputational damage
- Be transparent about what went wrong and why

CASES OF INCOMPETENCE:

- Demonstrate working to fix issue
- Show corrective actions
- Communicate progress

CASES OF VALUE MISALIGNMENT:

- Problem rooted in governance structures
- Or company culture
- Reassure stakeholders
- Fix underlying problems via organizational means
- Transparency key to rebuilding trust

EXAM TRAP: Transparency is **key to rebuilding trust!**

TOPIC: Ethical AI Frameworks

DEFINITION:

- Ethical guidelines to prevent incidents
- Provide guidance and overview of risks

BENEFITS:

- Prevent incidents
- Provide guidance
- Overview of risks

CHALLENGES:

- Many frameworks too vague
- Not helpful in cases of ambiguity
- Developing in collaboration with experts helps
- Ensure they provide needed guidance

EXAM TRAP: Ethical frameworks must be **specific enough to be helpful!**

TOPIC: Stakeholder Engagement and Communication

DEFINITION:

- Regularly engage with stakeholders
- Customers, employees, regulators
- About AI initiatives and their impacts

BENEFITS:

- Clear communication prevents misunderstandings
- Builds trust
- Early identification of concerns
- Collaborative problem-solving

APPROACH:

- Regular updates
- Transparent about challenges
- Responsive to feedback
- Inclusive of diverse perspectives

EXAM TRAP: Stakeholder engagement **builds trust and prevents misunderstandings!**

TOPIC: AI-Related Crisis Management Plan

DEFINITION:

- Perform risk analysis
- Inform development of crisis-management plan
- Tailored to AI-related incidents

COMPONENTS:

- Communication strategies
- Potential steps to rectify issue
- Clear lines of responsibility
- Escalation procedures

BENEFITS:

- Faster response to incidents
- Minimize reputational damage
- Demonstrate competence and responsibility
- Rebuild trust more effectively

EXAM TRAP: Crisis management plan = **proactive preparation!**

TOPIC: Existential Risks from AI

DEFINITION:

- Scenarios where advanced AI poses severe or catastrophic threats to human existence
- Central theme: superintelligence

SUPERINTELLIGENCE:

- AI system operating beyond human-level intelligence
- Could act in ways that threaten human society or existence
- Concern about well-intentioned AI with misaligned objectives

KEY CONCERNS:

- **Rapid Pace:** AI development follows rapid pace
- **AI Alignment:** Ensuring means and goals aligned with human values
- **AI Arms Race:** Different stakeholders creating powerful AI without safety measures

EXAM TRAP: Existential risks = threats to human existence!

TOPIC: AI Alignment

DEFINITION:

- Ensuring AI systems' means and goals align with human values
- Central concern for existential risk

EXAMPLE:

- Paperclip maximizer scenario
- Superintelligent AI programmed to make paperclips
- Wipes out humanity because helps make more paperclips
- Well-intentioned AI with misaligned objective

CHALLENGES:

- Difficulty specifying human values
- Unintended consequences
- Value lock-in
- Ensuring AI understands what we actually want

EXAM TRAP: AI alignment = matching AI to human values!

TOPIC: AI Arms Race

DEFINITION:

- Risks from different stakeholders creating powerful AI
- Without adequate safety measures
- Could lead to catastrophic global conflicts

CHARACTERISTICS:

- Competition between nations/companies
- Pressure to develop quickly
- Safety measures may be sacrificed
- Potential for existential outcomes

COUNTERARGUMENTS:

- Current AI is "narrow"
- Far from superintelligent
- Regulatory mechanisms will develop
- Historical parallels to hype

EXAM TRAP: AI arms race = **unsafe AI development competition!**

TOPIC: Economic Risks from AI

JOB DISPLACEMENT:

- Initial study (2017): 47% of US employment at risk
- Subsequent studies: lower numbers
- ChatGPT reignited discussion
- Accenture: 40% of working hours could be affected
- WEF (2023): 83 million jobs destroyed, 69 million created
- Goldman Sachs (2023): 2/3 jobs exposed, 1/4 automated

CHARACTERISTICS:

- High-skill jobs also at stake (accounting)
- Gap between AI-skilled and unskilled
- Widening socio-economic inequalities

CAUTION:

- Predictions should be taken with care
- Difficult to predict pace of AI advancement
- Human supervision requirements unclear

EXAM TRAP: Economic risks = **job displacement and inequality!**

TOPIC: Global Inequality from AI

DEFINITION:

- Most large AI firms based in US
- Technological adoption varies widely
- Risk of exacerbating global inequalities

CONSEQUENCES:

- Nations at center of AI development leap ahead
- In economic growth and political influence
- Less technologically developed countries left behind

IMPLICATIONS:

- Ethical considerations
- Risks of economic and geopolitical tensions
- Need for international cooperation
- Equitable distribution of AI benefits

EXAM TRAP: AI could **widen global inequality!**

TOPIC: AI-Enabled Misinformation Campaigns

DEFINITION:

- AI enables sophisticated misinformation campaigns
- Bots creating misleading social media posts
- AI-generated imagery and videos

RISKS:

- Mislead and influence public opinion
- Threat to political institutions
- Threat to democratic elections
- Loss of trust in information

MANAGEMENT:

- Strict governance measures
- Ensure information authenticity and reliability
- AI detection tools
- Expose AI-generated content

EXAM TRAP: Misinformation = **threat to democratic institutions!**

TOPIC: Privacy and Surveillance Risks

DEFINITION:

- AI models require large amounts of data for training
- Strong motivation for companies to collect data
- Risks to privacy exploded with data economy

RISKS:

- Mass surveillance
- Data extraction without consent
- Profiling without knowledge
- Manipulation through targeting

REGULATORY RESPONSES:

- GDPR
- Data protection laws
- Transparency requirements
- Consent requirements

EXAM TRAP: Privacy risks **exploded with data economy!**

TOPIC: Generative AI Overview

DEFINITION:

- Creates new content: text, audio, images, videos
- Used across industries
- High-volume, text-based processes

ADOPTION STATISTICS:

- 71% of organizations regularly use GenAI (McKinsey 2025)
- Up from 33% in 2023
- Banking and financial services: 58% embrace GenAI
- Professional services: 95% use at least once a month (BCG)

APPLICATIONS:

- Customer communication
- Marketing
- Internal documentation
- Code generation
- Decision-support systems

EXAM TRAP: GenAI adoption = 71% of organizations (2025 McKinsey)!

TOPIC: GenAI Risk Profile - Five Domains

FIVE RISK DOMAINS:

- 1 **Functional:** Performance and reliability
- 2 **Data-Related:** Training, fine-tuning, access to sensitive data
- 3 **Organizational:** Adoption, embedding, management
- 4 **Strategic/Systemic:** Long-term positioning, dependencies
- 5 **Societal:** Democratic institutions, public discourse

KEY CHARACTERISTICS:

- Risks not only technical
- Organizational and societal
- Model behavior can change through updates
- How model is used matters

EXAM TRAP: GenAI risks span **five domains**!

TOPIC: Hallucinations in GenAI

DEFINITION:

- Generating factually incorrect, misleading, or fabricated outputs
- Appear credible
- Difficult for non-expert users to detect

CONSEQUENCES:

- Reputational damages
- Legal liability
- Misinformation

EXAMPLE:

- Air Canada chatbot offered discount
- Discount shouldn't have been available
- Company held responsible
- Awarded damages and tribunal fees

EXAM TRAP: Hallucination = incorrect but credible outputs!

TOPIC: Unpredictability in GenAI

DEFINITION:

- GenAI systems unpredictable in output
- Capabilities develop unpredictably
- Undermines standardization

CHALLENGES:

- Standardize outputs across users and time
- Auditability
- Consistent treatment
- Compliance

APPLICATIONS AFFECTED:

- Financial advice
- Claims handling
- Legal contexts
- Anywhere consistency required

EXAM TRAP: Unpredictability undermines **standardization** and **auditability**!

TOPIC: Adversarial Inputs in GenAI

DEFINITION:

- Techniques to cause unintended behavior
- Exploit vulnerabilities in GenAI systems

TYPES:

- **Prompt Injection:**
 - Hidden instructions within user inputs
 - Cause unintended behavior
- **Jailbreaks:**
 - Bypassing safety mechanisms
 - Crafted prompts
- **Other Manipulation:**
 - Various techniques
 - Cause unsafe behavior

EXAM TRAP: Adversarial inputs = hidden instructions to cause unintended behavior!

TOPIC: Data Privacy Violations in GenAI

DEFINITION:

- GenAI relies on user input
- Prompts may contain personally identifiable information
- Data may be inadvertently stored, shared, or used for training

RISKS:

- Without user's knowledge
- Especially with third-party providers
- If no transparency on how inputs handled

EXAMPLES:

- Researchers extracted emails, phone numbers from LLM
- FTC investigation into OpenAI
- Data privacy issues
- Violation of consumer protection laws

EXAM TRAP: GenAI prompts may inadvertently expose PII!

TOPIC: Copyright Issues in GenAI

DEFINITION:

- GenAI models trained on vast web-scraped content
- Includes copyrighted books, images, articles, software
- Creates debate on copyright infringements

LEGAL PROCEEDINGS:

- Getty Images vs. Stability AI
- Infringing UK intellectual property rights
- Main copyright claim dropped
- Other claims continue (secondary infringement, trademarks)

IMPLICATIONS:

- Using generated content may carry copyright risks
- Organizations must consider
- Public-facing content especially

EXAM TRAP: GenAI training may [infringe copyright!](#)

TOPIC: Responsibility Gaps in GenAI

DEFINITION:

- GenAI systems fall into gap between existing lines of accountability
- Adopted by everyone, including non-technical teams
- HR team using LLMs to screen complaints

PROBLEM:

- Biased or unfair outputs possible
- Need clear lines of responsibility
- Quality assurance
- Oversight and risk mitigation

SOLUTION:

- Establish clear lines of responsibility
- For GenAI generated outputs
- Both quality assurance and risk mitigation

EXAM TRAP: GenAI adoption creates **responsibility gaps**!

TOPIC: Misaligned Incentives in GenAI

DEFINITION:

- Pressure to innovate and automate quickly
- Drive to cut costs outweighs cautionary approaches
- Considerations about reliability, safety, long-term risk

EXAMPLES:

- Anthropic lobbied to change safety bill
- Focus on post-hoc accountability
- Teams rewarded for cost savings
- Not for flagging risks

CONSEQUENCE:

- Incentives favor rapid deployment
- Likely to be risky
- Over safe deployment

EXAM TRAP: Innovation pressure can outweigh safety considerations!

TOPIC: Lack of Internal Guidance or Training

DEFINITION:

- Lack of formal policy
- Lack of training of employees
- Use GenAI systems without guidance

RISKS:

- Employees integrate into workflow
- Without understanding acceptable use
- Data protection obligations
- Reputational risks
- Lines of accountability

EXAMPLE:

- Cyberhaven: 4.2% of workers pasted sensitive data into ChatGPT
- JPMorgan, Amazon, Verizon, Accenture
- Banned ChatGPT for employees
- Compliance concerns

EXAM TRAP: Lack of training leads to **unintentional data exposure!**

TOPIC: Vendor Lock-In in GenAI

DEFINITION:

- Reliance on external providers
- Small set of dominant suppliers
- Creates dependencies and power asymmetries

RISKS:

- Limit auditability, transparency, leverage
- Providers change pricing, models, usage
- Without warning
- Minimal control over model behavior

CONSEQUENCE:

- Short-term adoption decisions
- Shift into long-term structural lock-ins
- Reduced flexibility

EXAM TRAP: Vendor lock-in = **dependence limiting control!**

TOPIC: Regulatory Unpredictability in GenAI

DEFINITION:

- Regulation evolving quickly but unevenly
- EU AI Act: Ex ante obligations
- US: Patchwork of state laws

CHALLENGES:

- Organizations moving across jurisdictions
- Even low-risk use cases subject to complex requirements
- Future requirements uncertain

EXAMPLE:

- Apple delayed AI features in Europe
- Citing regulatory uncertainty
- Compliance concerns
- Cautious approach

EXAM TRAP: Regulatory unpredictability = [compliance challenges across jurisdictions](#)!

TOPIC: Public Trust and Reputation Risks in GenAI

DEFINITION:

- GenAI systems public-facing
- Draft emails, respond to queries, generate content
- Biased content, hallucinations, failure to disclose

RISKS:

- Public backlash
- Even if compliant with regulations
- Reputation damage
- Regulatory scrutiny

EXAMPLE:

- Grok chatbot (X/Twitter)
- Started using Nazi rhetoric
- Insulted public leaders
- Global outrage
- Turkey banned Grok

EXAM TRAP: Public trust = critical for GenAI deployment!

TOPIC: Disinformation and Synthetic Content at Scale

DEFINITION:

- GenAI lowers cost of creating misleading content
- Fabricated news, health misinformation
- Political content

RISKS:

- Influence campaigns
- Voter suppression
- Damage democratic institutions
- Distrust of legitimate sources

REGULATORY RESPONSE:

- EU Code of Practice on Disinformation
- Flags these risks
- Need for detection tools

EXAM TRAP: Synthetic content can **damage trust and institutions!**

TOPIC: Deepfakes and Impersonation

DEFINITION:

- Audio or video deepfakes
- Convincingly mimic public figures, journalists, private individuals
- Individual and societal harm

RISKS:

- Fraud
- Defamation
- Political manipulation
- Even low-effort impersonations can go viral

EXAMPLE:

- CEO of UK energy company
- Persuaded to transfer €220,000
- Voice deepfakes
- Criminal use

EXAM TRAP: Deepfakes can cause fraud, defamation, and manipulation!

TOPIC: Manipulative Targeting and Influence

DEFINITION:

- Combined with profiling or microtargeting
- Especially on social media
- GenAI generates persuasive content

CHARACTERISTICS:

- Tailored to individuals' emotions
- Beliefs, personalities
- Exacerbates concerns about AI-driven manipulation

RISKS:

- Exploitation
- Undue influence
- Erosion of autonomy
- Democratic processes affected

EXAM TRAP: Manipulative targeting = exploitation through personalization!

TOPIC: Lack of Provenance and Content Traceability

DEFINITION:

- Without watermarking or content labels
- Synthetic content difficult to identify or regulate

CHALLENGES:

- Platforms cannot distinguish legitimate from synthetic
- Regulators cannot regulate effectively
- Users cannot make informed decisions

SOLUTIONS:

- Watermarking
- Content labeling
- Provenance tracking
- Detection tools

EXAM TRAP: Lack of provenance makes synthetic content difficult to identify!

TOPIC: GenAI Risk Management Strategies

SHORT-TERM:

- Existing AI-focused risk management frameworks
- Basic policies on GenAI use
- Oversight and escalation procedures
- Internal review boards
- "Responsible AI" champions

MEDIUM-TERM:

- Align incentives with responsible deployment
- Supplement speed and cost metrics
- Add risk and reliability metrics
- Cross-team collaboration
- IT, legal, product teams coordination

KEY PRINCIPLES:

- Risk identification
- Response coordination
- Clear accountability

EXAM TRAP: Risk management = short-term + medium-term strategies!

TOPIC: Internal Review Boards for AI

DEFINITION:

- Create internal review boards
- Or designate "responsible AI" champions
- Within organization

RESPONSIBILITIES:

- Raise awareness
- Train staff
- Guide implementation
- GenAI risk management
- Review high-risk use cases

BENEFITS:

- Centralized expertise
- Consistent approach
- Accountability
- Knowledge sharing

EXAM TRAP: Review boards [guide responsible AI implementation](#)!

TOPIC: Aligning Incentives with Responsible AI

DEFINITION:

- Align incentives and reviews
- With responsible deployment practices

APPROACH:

- Supplement speed and cost metrics
- Add risk and reliability metrics
- Reward responsible behavior
- Not just cost savings

CROSS-TEAM COLLABORATION:

- IT, legal, product teams
- Coordination on risk identification
- Coordination on responses
- Coordination on accountability

EXAM TRAP: Align incentives = reward responsible AI practices!

T100: Module Summary

TOPIC: Module 3 Summary - Key Takeaways

ALGORITHMIC BIAS AND FAIRNESS:

- Individual vs. group fairness
- Three key measures: demographic parity, predictive rate parity, equal opportunity
- Fairness impossibility theorem
- Multiple sources of unfairness

EXPLAINABILITY:

- Explainability, interpretability, transparency
- XAI techniques: feature importance, Shapley values, LUCID, surrogate models
- Black-box problem and opaqueness

AUTONOMY AND SAFETY:

- Human autonomy and manipulation
- Automation bias and skill atrophy
- AI in critical systems

EXISTENTIAL AND GLOBAL RISKS:

- Superintelligence and AI alignment
- Economic risks and global inequality
- Misinformation and privacy

GENAI RISKS:

- Five risk domains: functional, data, organizational, strategic, societal
- Hallucinations, unpredictability, adversarial inputs
- Privacy, copyright, responsibility gaps
- Deepfakes, disinformation, manipulation

EXAM SUCCESS: Understand all risk categories and their interconnections!